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LUNG RETINANET: Lung Cancer Prediction and Classification

By

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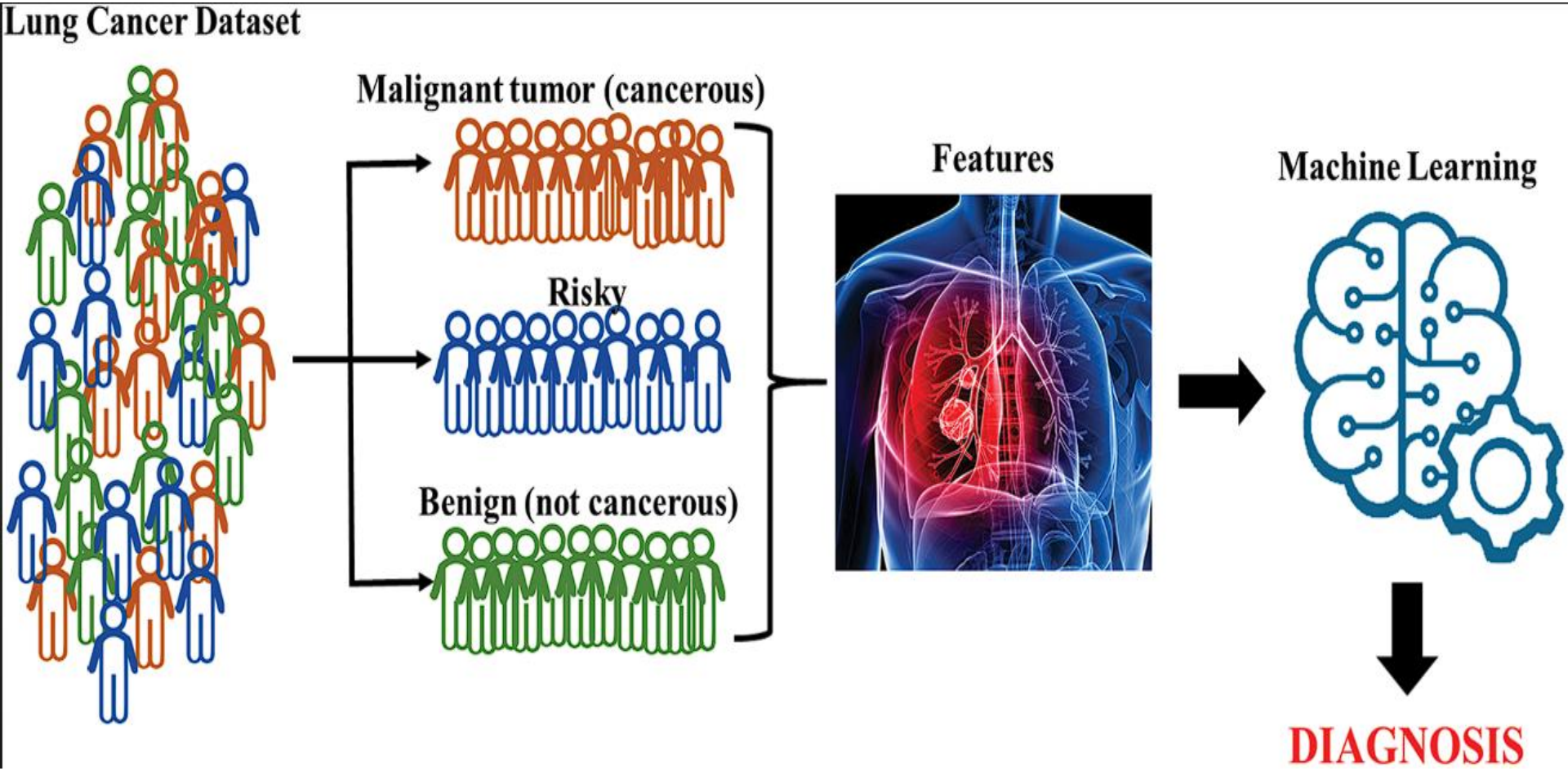
Abstract

- In the contemporary era, both smokers and non-smokers were known to have tumour cells in their lungs.
- It is noted that tumour cell prognosis at baseline is a challenge for any radiologist.
- The simple system consists of Data Collection, Feature Extraction, Building Model and locating of lung tumors.
- RCNN, CNN, Mask RCNN network based detector to provide statics for proposed model.
- ResNet 101 is used as backbone consisting of Feature Fusion Model and Dilated Context Module to achieve classification and regression subnetwork.
- The proposed model is on Deep Neural Network which gives 98.6% accuracy results and is an extensible and lightweight algorithm providing flexible GUI.

Introduction

- The lungs are two soft organs in chest of any living being that regulate respiration.
- If cell division in the lungs goes unchecked it forms solid tumors or lymph nodes with abnormal cells that damages normal body tissues and these cancer cells spread throughout our body.
- In India, it is estimated that cancer will increase by 12.8% in the year 2025.
- The recognition of cancer starts with imaging tests include X-rays, MRI, CT and positron emission tomography, sputum cytology, lung cancer stages are 1 to 4.
- The proposed system have improved in deep learning, computer assisted machine learning identification system.
- ResNet 101 (Residual Network) deposit of the pre-trained network is trained on more than 700 images from Image Net Database which is then classified into two classes here for Benign and Malignant tumours in lungs.

Introduction



PROBLEM STATEMENT

- The challenge in diagnosing lung cancer are the manual labor experienced in medical field, when a less experienced or inexperienced person fails to provide treatment and sometimes gives false prognosis.

Objectives

- To research and clinical setting for lung cancer prediction purpose.
- To Provide early detection and classify disease based on type of tumors.
- To increases the accuracy of analysis also identifies specific boundaries of tumors.
- To provide improved facilities of cancer detection in remote areas.

Literature Review

Paper Title	Author	Summary	Limitations
1.Evaluation of Machine Learning methods for Lung Cancer Prediction	John Doe; Jane Smith.	ML applications in lung cancer prediction including feature selection, classification algorithm and data pre-processing technique.	It consumes more training time. Accuracy is less than 90%.
2. Deep learning methods with contextual feature fusion modules for lung cancer prediction	James Anderson; Sophia Martinez	It discusses ensemble methods for integrating multi-omics data, transferring learning, and improving prediction accuracy.	Deep learning approach consumes more training time. Accuracy in detection is less than 75%.
3. Residual Convolutional Neural Network for Predicting Patient Survival in Lung Cancer using RNA Data	Zhao; Zhongming, et al.	This paper introduces a ResCNN forpredicting patient survival in lung cancer using RNA-seq data. The study demonstrates the effectiveness of ResNet-based architectures in survival prediction tasks.	Test evaluation for only predicting survival rate. Do not recommend medicines

Literature Review

Paper Title	Author	Summary	Limitations
4. ML methods for lung cancer detection and diagnosis	Alice Johnson; Michael Brown	It focuses on recent advances in machine learning techniques for lung cancer detection and diagnosis. It covers Python library uses such as scikit-learn and TensorFlow for predictive model building.	Lack of transparency in ML Algorithm. Low reproducibility and interpretability of result.
5. Lung Cancer Detection and Classification with 3D Convolutional Neural Networks	Sun; Wenqing, et al.	This study explores the use of 3D CNNs for lung cancer detection and classification. Python and deep learning frameworks such as TensorFlow are employed for model training and evaluation.	Hinders real-time implementation and scalability. Detecting accuracy is variable.
6. DL methods for lung cancer detection	Emily Wang; David Chen	Use of deep learning methods in lung cancer prognosis. This includes CNNs, RNNs, and their variations. Python frameworks such as Keras and PyTorch are discussed for lung cancer prediction.	Low performance. Fairness in quality datasets.

Design Methodology

- The proposed system methodology consists of Feature Fusion module, Extended Presentation module(dilated context), RCNN, CNN, Masked R-CNN as shown in figure 1.

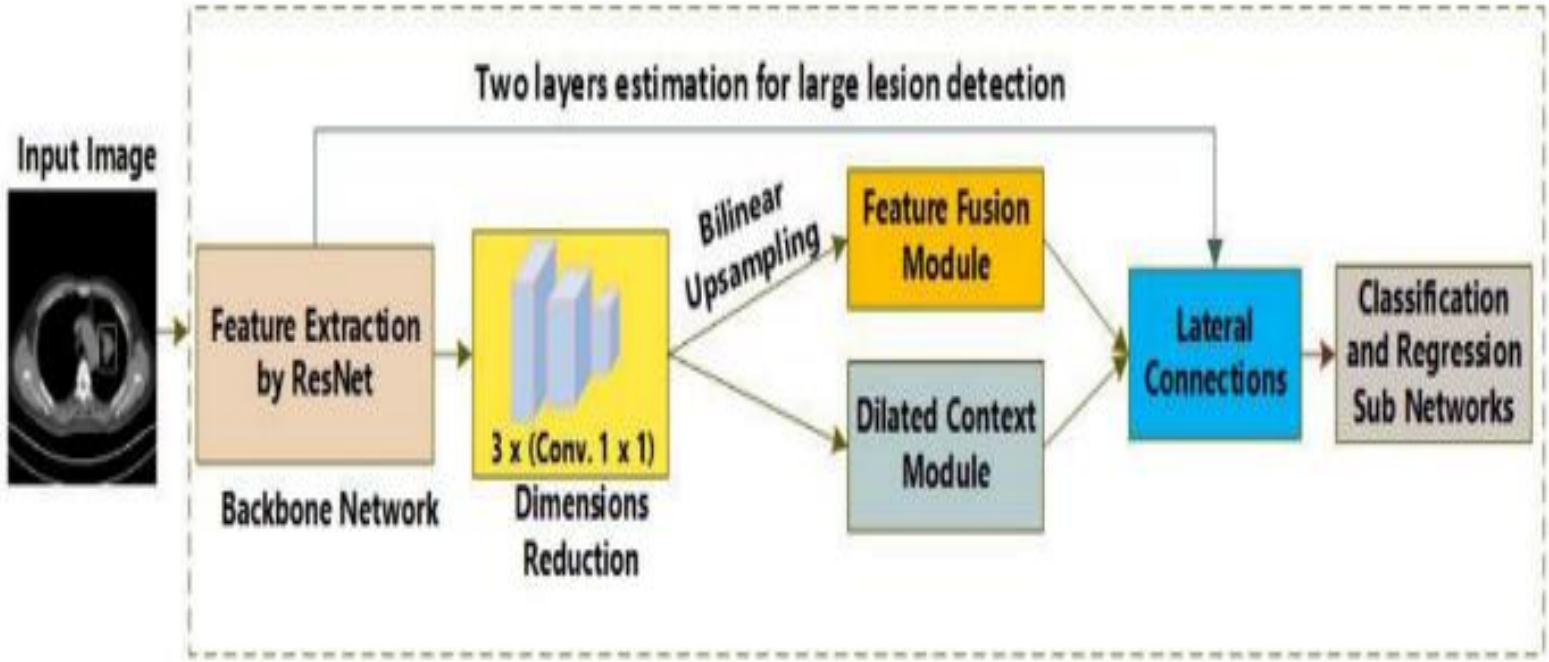


Figure 1: Architecture of Lung RetinaNet.

Design Methodology

❖ **Feature Fusion Module**

- It combines extractions at different levels of the network, providing a comprehensive prospect of the embedded images.
- It use functions such as joining, element-wise joining, or convolutional functions to efficiently combine features from multiple levels.

❖ **Extended Presentation Module**

- It increases the receptive field of the diffraction layers without increasing parameters or reducing spatial resolution.
- It capture large amounts of contextual information in the pixel of the input image, for understanding the structural relationships in different locality of the lung scan.

❖ **RCNN**

- Medical professionals annotate images to mark areas of interest, especially tissue suspected of cancer. This specification acts as a ground truth score for training the RCNN model.

Design Methodology

- It Performance metrics such as accuracy, sensitivity, and specificity are calculated to evaluate the efficiency of the model.

❖ **CNN**

- Important in predicting lung cancer, using their ability to identify surface features from medical images such as chest CT scans to accurately identify tumors.
- It includes data pre processing, model training and feature extraction are trained in clinical scenarios that can classify and process data.

❖ **Masked R-CNN**

- It offers improved capability not only for types of tumors, but also by defining their specific boundaries.
- It excels in detecting and accurately classifying pulmonary nodules in medical images such as chest CT scans and allows more exhaustive examination of the characteristics of the nodules, and improves the diagnosis.

Working Principles

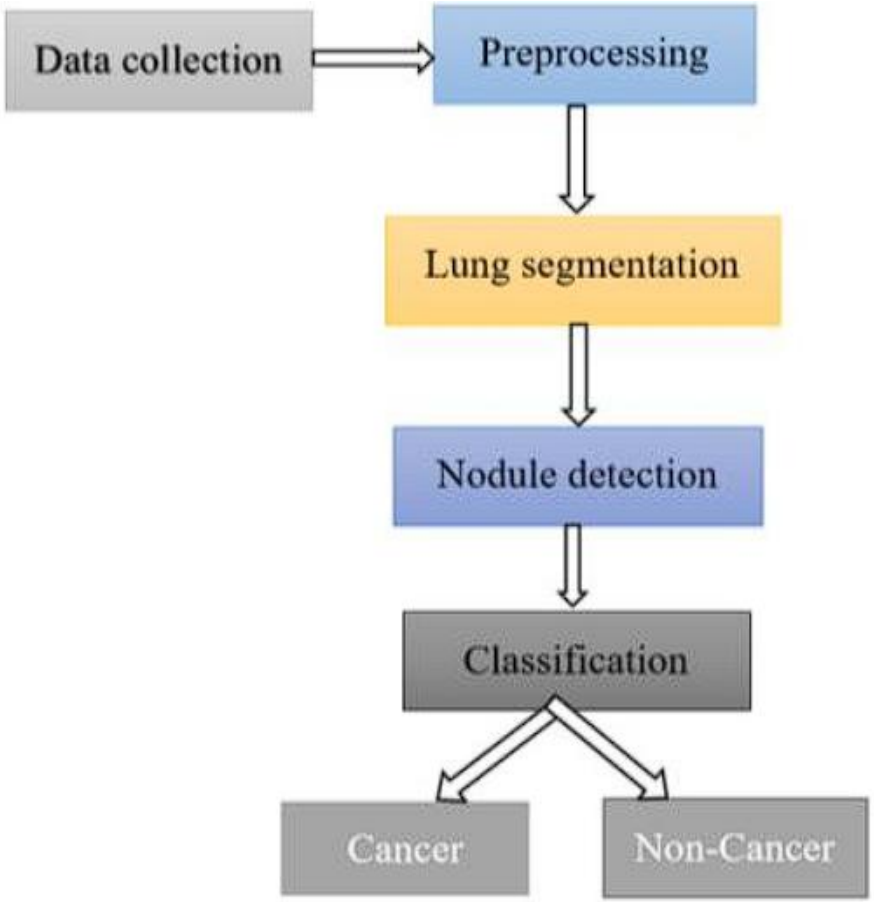


Figure 2: Dataflow of Implementation.

Working Principles

Data Collection:

- Collecting diverse and representative data from reliable sources is crucial. This includes medical databases, research repositories, and hospitals with consented patient data.
- Ensure the data encompasses various demographics, tumor characteristics, and treatment outcomes.

Data Preprocessing:

- This involves handling missing values, normalizing features, and encoding categorical variables.
- In medical datasets, dealing with imbalanced classes and noisy data is common and requires special attention.

Lung Segmentation:

- Lung segmentation involves delineating lung regions from medical images, typically CT scans, to isolate them for further analysis.
- Python offer tools for image preprocessing, segmentation, and visualization.

Working Principles

Nodule Detection:

- Nodule detection focuses on identifying and characterizing pulmonary nodules, potential indicators of lung cancer, within segmented lung regions.
- Training data typically consists of annotated CT scans, where nodules are labeled by radiologists. Once trained, the model can detect nodules in unseen CT images, aiding in early diagnosis and treatment planning.

Classification Module:

- The classification module employs machine learning algorithms to classify patients into two categories: those with lung cancer and those without.
- The classification model is trained on labeled data, where instances are associated with binary labels indicating the presence or absence of lung cancer.

Working Principles

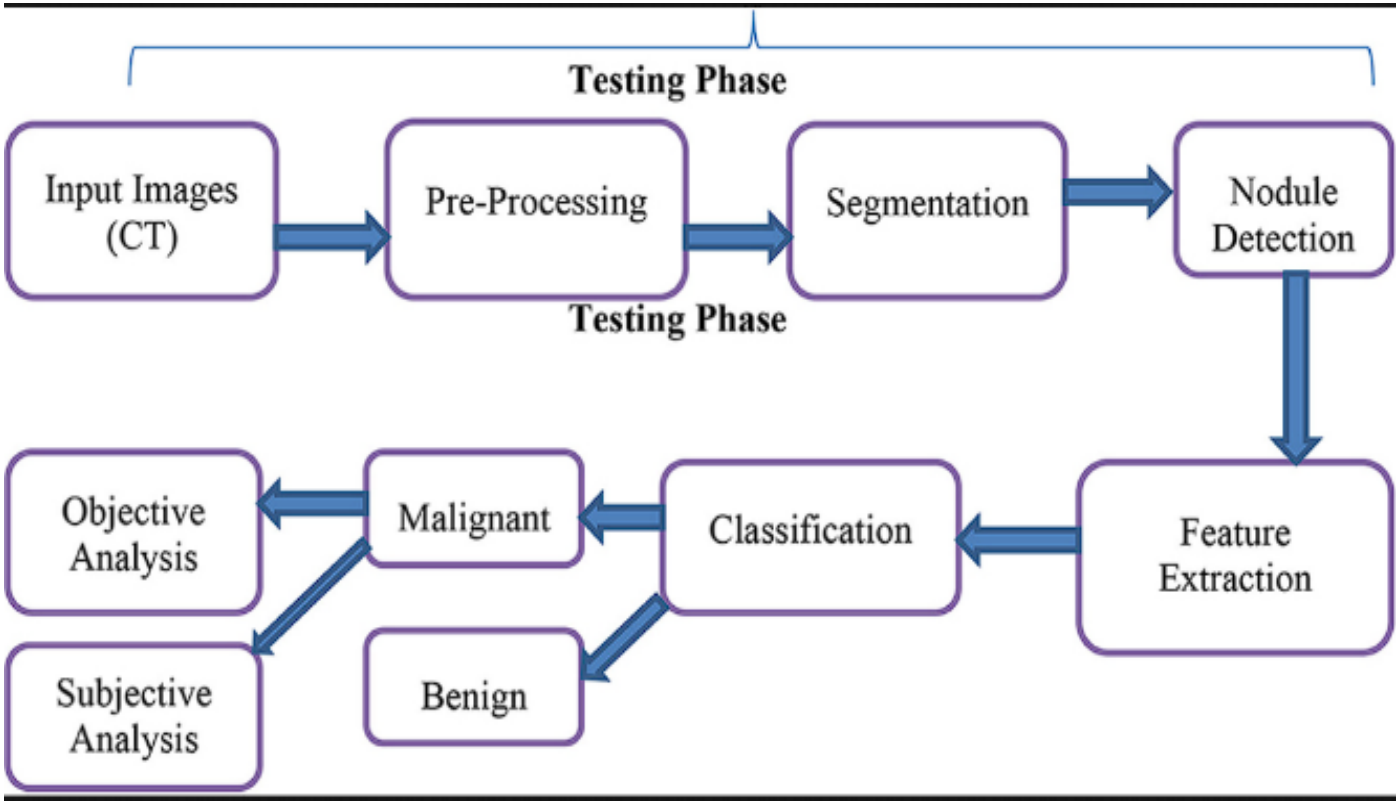


Figure 3: Workflow of the system.

Working Principles

- 750 samples are fed which includes both benign and malignant tumours to train our system Then we place a patient CT image to identify lung tumours.
- The ResNet 101 helps to extract all features from input with dilated context block fusion then categorization and regression subnetworks is placed and displayed which provides result.
- If patient is found with tumour cells of malignant or benign, then shows where it is located with objective analysis and subjective analysis as in figure 3

Result

- To get more information from the given CT image dilated context module provides best microscopic zoom views so that any of the sensitive information will never get missed even if their exist glitch in image and also locates the position within blue box as shown in figure 4.

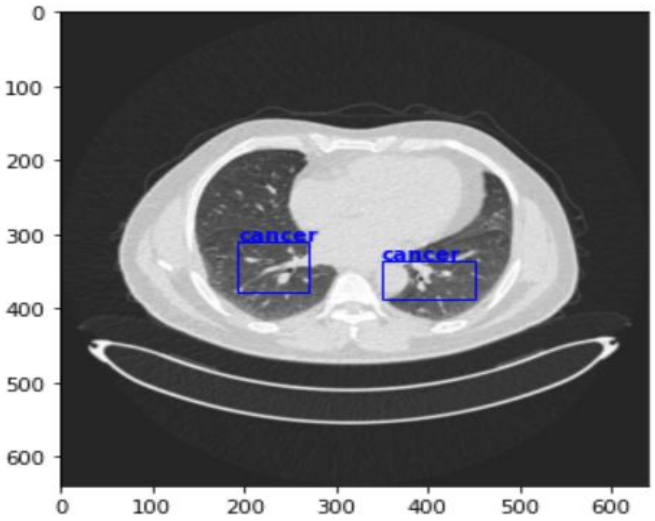


Figure 4: Location of tumour cells in lungs.

Result

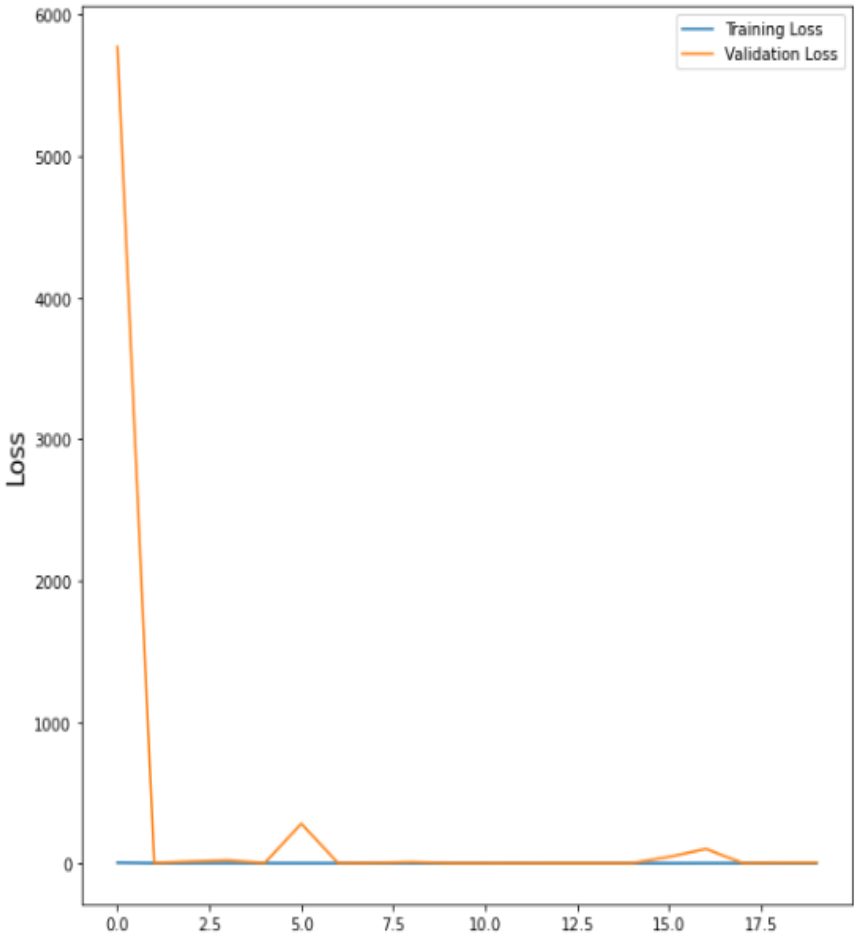


Figure 5.1: Graphical representation of training loss and validation loss.

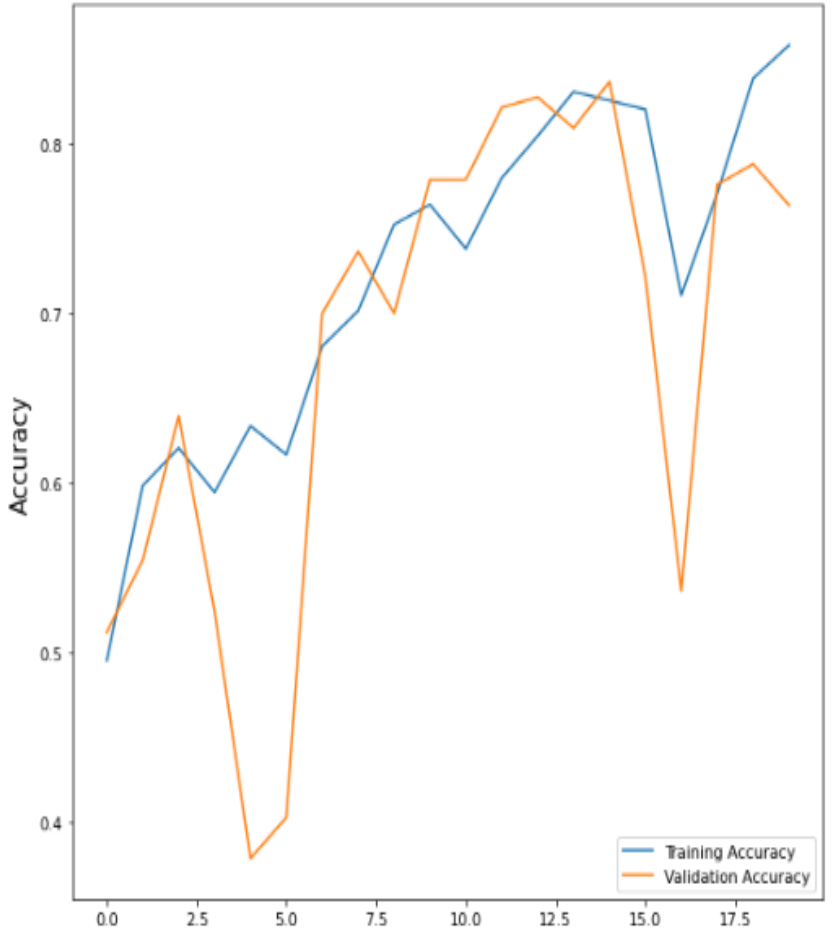


Figure 5.2: Graphical representation of Training accuracy and validation accuracy.

Conclusions

- The work provides 98.6% accuracy and robustness using two modules in our ResNet 101 consisting of Feature Fusion and Dilated Context Module with optimal aggregation and feature extraction from the image with the given reduced data set converted into useful data.

- It gives more information from the given CT image dilated context module provides best microscopic zoom views so that any of the sensitive information will never get missed even if their exist glitch in image.

- Research and improved performance although may acquire only a few difficulties due to resolutions of images but it works well.

References

[1]"Deep Learning-Based Detection of Lung Cancer on CT Scans: Comparison of 2D and 3D Convolutional Neural Networks" by Ardila, Diego, et al. 2019.
[2]"Residual Convolutional Neural Network for Predicting Patient Survival in Lung Cancer using RNA Data" by Zhao, Zhongming, et al. 2017.

[3]"Lung Cancer Detection using Deep Learning Technique ResNet“ by Sheikh, Zeeshan Rasheed, et al. 2020.
[4]"Residual Neural Network for Early Diagnosis of Lung Cancer on CT Images" by Zhang, Xiao, et al. 2018.

[5]"Lung Cancer Detection and Classification using Convolutional Neural Networks: A Comprehensive Study" by Shrivastava, Alok, et al. 2021.
[6]"Deep Learning-Based Classification of Lung Cancer Histopathology Images using Transfer Learning" by Bayramoglu, Neslihan, et al. 2016.

[7]"Predicting Survival Time of Lung Cancer Patients using Radiomic Analysis" by Yang, Jun, et al. 2018.
[8]"Deep Learning-Based Automated Detection of Lung Cancer with CT Imaging: A Review" Authors: Chen, Houfu, et al. 2020.

[9]"Lung Cancer Detection and Classification with 3D Convolutional Neural Networks" Authors: Sun, Wenqing, et al. 2019.